

Engineering Physics (2025)
Course code 25PY101
Module 2 Unit 2: Optoelectronics

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M2U2 Plan

- 1 p-n junction diode – forward and reverse conditions
- 2 Solar cell
- 3 Photo-diode
- 4 Direct and indirect band gap semiconductors
- 5 Light Emitting Diode

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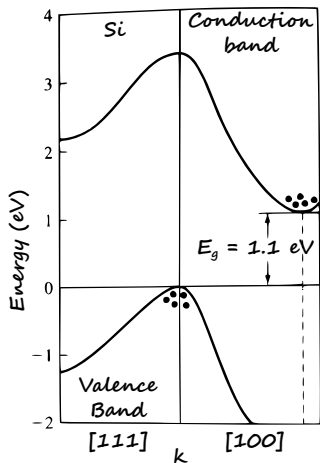
Light Emitting Diode

Definition

Light Emitting Diode (LED) is the opto-electronic device based on radiative recombination of charge carriers.

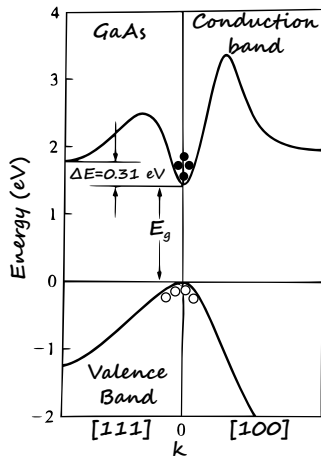
- LED is based on pn junction diode under forward bias.
- Under forward bias, the electrons and holes crossing the depletion region undergo radiative recombination leading to generation of light.
- It is useful to use a semiconductor with a direct band gap semiconductor.

Elemental semiconductors – Si



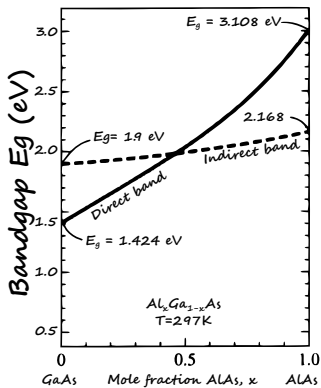
- Si is an elemental semiconductor.
- It is an indirect band gap semiconductor.
- The valence band maximum and the conduction band minimum occur at different values of wavevector.

Binary semiconductors – GaAs



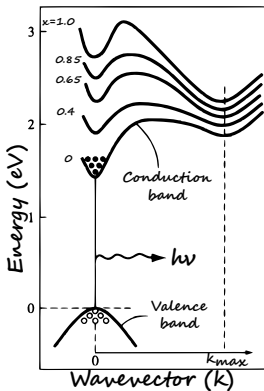
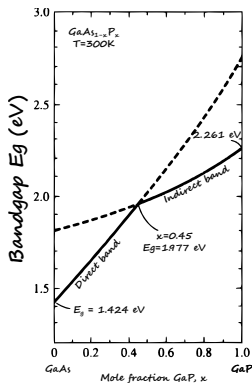
- GaAs is a binary semiconductor.
- It is a direct band gap semiconductor.
- The valence band maximum and the conduction band minimum occur at same value of wavevector $k = 0$.

Ternary semiconductors – $\text{Al}_x\text{Ga}_{1-x}\text{As}$



- $\text{Al}_x\text{Ga}_{1-x}\text{As}$ is a ternary semiconductor.
- It can be considered as a binary alloy of GaAs and AlAs. If the mole fraction of AlAs is considered as x and mole fraction of GaAs as $1 - x$, then the chemical formula is $\text{Al}_x\text{Ga}_{1-x}\text{As}$. Therefore $x = 0$ corresponds to pure GaAs whereas $x = 1$ corresponds to pure AlAs.
- From $x = 0$ to $x = 0.45$, the alloy is a direct band gap material. From $x = 0.45$ to $x = 1$, the alloy is an indirect band gap material.
- In the energy band gap vs mole fraction, there is a cross over from direct band gap to indirect band gap at $x = 0.45$.

Ternary semiconductors – $\text{GaAs}_{1-x}\text{P}_x$



- $\text{GaAs}_{1-x}\text{P}_x$ is a ternary semiconductor with mole fraction x of GaAs and mole fraction $1 - x$ of GaP.
- From $x = 0$ to $x = 0.45$, the alloy is a direct band gap material. From $x = 0.45$ to $x = 1$, the alloy is an indirect band gap material.
- In the bandstructure, the conduction band shifts from $k = 0$ to $k = k_{\text{max}}$ as the mole fraction of GaP crosses $x = 0.45$.

Summary of opto-electronic devices

- From the perspective of bias

Reverse bias	Zero bias	Forward bias
Photo-diode	Solar cell	LED

- From the perspective of depletion region width

Larger width	Nominal width	Smaller width
Low current	Nominal current	Large current

- From the perspective of power consumption

Low Power	Nominal Power	High Power
Consumption	Generation	Consumption

- From the perspective of capacitance

Low Capacitance	Nominal Capacitance	High Capacitance
High frequency	–	–